

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem Solving Task

Code of Conduct:

I Abangjithan S / 192521299 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S. A. B.

Annexure A

Industry Problem Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.
2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is

typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable

to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to

floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for

security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution

Star topology - 20 Person startup

→ Star topology is suitable for a 20-person startup because it is simple and easy to manage.

→ All computers and devices are connected to a central switch.

→ A 24-Port managed switch can be used to connect all office devices.

→ Cat6 UTP cables can be used to connect computers to the switch.

→ A route / firewall is connected to the switch for internet access and security.

→ A wifi access point can provide wireless connectivity for laptops & mobiles.

→ If one cable fails, only that particular computer is affected.

→ Other computers can continue working without interruption.

→ New computers can easily be added to the network when the company grows.

→ Therefore, star topology is reliable, simple and easy to maintain.

2. Mesh Topology - Banks

→ A bank requires high reliability and minimum network downtime.

→ Therefore, mesh topology is suitable for banking networks.

→ Important routers, switches and servers can have multiple connections.

→ If one connection fails, data can travel through another available path.

→ A partial mesh can be used for the main / core network.

→ Redundant routers and switches can provide backup connections.

→ Fiber-optic cables high bandwidth and low latency.

→ Fiber cables can be used for high speed network communication.

→ Protocols such as OSPF or BGP can automatically find alternate paths.

→ Therefore, mesh topology provides high reliability and fault tolerance.

Bus topology - classroom lab

→ Bus topology can be used for small classroom with a limited budget.

→ All computers share a single common communication cable.

→ Traditional bus topology uses coaxial cable as the main backbone.

→ T-connectors and BNC connectors are used to connect computers.

→ Terminators are placed at both ends of the cable to prevent signal reflection.

→ This topology requires less networking hardware.

→ It is suitable for light activities such as browsing and file sharing.

→ However, heavy network traffic may cause collisions.

→ If the main cable fails, all connected computers can be affected.

→ Therefore, bus topology is a simple and low-cost solution for small labs.

4. Hybrid topology - University campus

→ A university campus has many buildings and thousands of users.

→ Therefore, a hybrid topology is suitable for the campus network.

→ Different topologies can be combined according to different requirements.

→ A fiber-based ring or mesh can be used for the campus backbone.

→ Star topology can be used inside individual buildings.

→ Fiber-optic cable can connect different buildings at high speed.

→ Cat 6 cables can connect computers to access switches.

→ WiFi access points can provide wireless connectivity throughout the campus.

→ The network can easily be expanded when new buildings are added.

→ Therefore, hybrid topology provides scalability, reliability and easy management.

5. Multiple department company Network

→ A company may have departments such as HR, Finance, Sales and IT.

→ Each department can be connected using separate layer 2 switches.

→ The department switches are connected to a central layer 3 switch or routers.

→ This allow communication between different departments.

→ separate VLANs can be created for each department.

→ VLANs improve security and reduce unnecessary broadcast traffic.

→ Inter-VLAN routing allows controlled communication between departments.

→ A router / firewall can provide internet access and network security.

→ Redundant links can be added to improve network reliability.

→ Therefore, this design provides security, performance and easy expansion.